

7 (a)

- (i) In this technique, particles of electron exhibit wave-like behaviour as evidenced by the diffraction pattern produced by the electron beam.
- (ii) In order for diffraction to occur, the associated wavelength of the electrons is about 0.1 nm.

By de Broglie equation,

$$\text{momentum of the electrons, } p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{0.1 \times 10^{-9}} = 6.63 \times 10^{-24} \text{ kg m s}^{-1}$$

Kinetic energy of the electrons,

$$KE = \frac{p^2}{2m} = \frac{(6.63 \times 10^{-24})^2}{2(9.11 \times 10^{-31})} = 2.4 \times 10^{-17} \text{ J} \approx 2 \times 10^{-17} \text{ J}$$

(iii) Momentum, $p = mv$

$$\Rightarrow \frac{\Delta p}{p} = \frac{\Delta v}{v} = 10\%$$

Uncertainty in electron's momentum,

$$\Delta p = \frac{10}{100} (6.63 \times 10^{-24}) = 6.63 \times 10^{-25} \text{ kg m s}^{-1}$$

Using Heisenberg's uncertainty principle, $\Delta x \Delta p \geq h$

$$\text{uncertainty in electron's position, } \Delta x \geq \frac{h}{\Delta p} = \frac{6.63 \times 10^{-34}}{6.63 \times 10^{-25}} \approx 1 \times 10^{-9} \text{ m}$$

(iv) Energy of photon,

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}) (3.0 \times 10^8)}{0.1 \times 10^{-9}} = 2.0 \times 10^{-15} \text{ J} \approx 2 \times 10^{-15} \text{ J}$$

(b)

(i) Using $KE = \frac{p^2}{2m}$

momentum of the protons,

$$p = \sqrt{2m(KE)} = \sqrt{2(1.67 \times 10^{-27})(2.4 \times 10^{-17})} = 2.8 \times 10^{-22} \text{ kg m s}^{-1}$$

$$\text{associated wavelength of the protons, } \lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{2.8 \times 10^{-22}} \approx 2 \times 10^{-12} \text{ m}$$

(ii) The use of protons would make a *more* effective probe because its associated wavelength is *smaller* than that of the electrons, thus producing images of higher resolution.

Section B

8(a)(i) The loss in GPE of the ball is converted into its KE